

**Asset Decumulation in Retirement:  
Patterns, Predictors, and the Role of Financial Literacy**

June 4, 2025

Hessam Bavafa, Anita Mukherjee, Tyler Q. Welch

Wisconsin School of Business, University of Wisconsin-Madison

***Abstract***

We study how U.S. retirees draw down non-housing financial assets using longitudinal data from the Health and Retirement Study (1995 to 2020). While lifecycle models predict gradual decumulation to support consumption, we find that many retirees retain substantial wealth into later life. IRA balances decline steadily, consistent with required minimum distribution rules. Stock assets are decumulated more quickly, especially among middle-wealth households. In contrast, checking and savings balances remain stable. Financial literacy is strongly associated with slower decumulation, higher asset levels, and reduced debt. These patterns suggest that asset type, institutional rules, and individual capabilities jointly shape financial outcomes in retirement.

*Acknowledgments:* This research was supported by a grant from the TIAA Institute. E-mails of authors, in order: bavafa@wisc.edu, anita.mukherjee@wisc.edu, tqwelch@wisc.edu. Mailing address for Anita Mukherjee, corresponding author: 5279 Grainger Hall, 975 University Ave, Madison, WI 53706. The views expressed in this paper are those of the authors and do not necessarily reflect those of the Institute. We are grateful to Brent J. Davis for helpful comments and to Hoyoung Yoo for research assistance.

## 1. Introduction

The shift from wealth accumulation to decumulation in retirement represents a critical but understudied phase of the financial lifecycle. Standard life-cycle consumption models predict that individuals should draw down assets gradually to smooth consumption over an uncertain horizon (Yaari 1965). In practice, however, many retirees retain substantial financial wealth well into later life, deviating from model predictions in ways that are not easily explained by precautionary savings or bequest motives alone (Christensen, Kallestrup-Lamb, and Kennan 2022; De Nardi et al. 2025; Lockwood 2012). These persistent deviations from theoretical benchmarks have been referred to as the “retirement-consumption puzzle” (Banks, Blundell, and Tanner 1998; Hurd and Rohwedder 2008; Olafsson and Pagel 2018).

This paper examines patterns of asset decumulation among U.S. retirees using data from the Health and Retirement Study (HRS) spanning 1995 to 2020. We focus on non-housing financial assets and track retirees over a 12-year period following age 65. Our analysis emphasizes heterogeneity across wealth distribution, asset categories, and levels of financial literacy. While prior work has documented broad patterns in wealth retention (Poterba, Venti, and Wise 2011; Brown, Poterba, and Richardson 2025), we contribute by decomposing decumulation across asset types and examining how institutional incentives, particularly those embedded in tax-advantaged retirement accounts, shape spending trajectories.

Economic theory often treats wealth as fungible, but institutional features of the retirement system and revealed preferences suggest otherwise. We distinguish between three broad categories of financial assets: IRAs and Keogh accounts (the latter of which are tax-deferred plans for those self-employed), taxable investment accounts such as stocks and mutual

funds, and liquid savings such as checking and money market accounts. IRAs are subject to Required Minimum Distributions (RMDs), which induce predictable drawdowns. Stock assets, although not subject to mandated withdrawals, are decumulated more rapidly, especially among middle-wealth households. This may reflect liquidity needs, tax considerations, or portfolio rebalancing. In contrast, balances in checking and savings accounts are more stable and often persist through retirement, even for individuals with limited overall wealth. These differences suggest that decumulation behavior depends not only on institutional rules but also on asset liquidity, mental accounting, and perceived risk.

Decumulation patterns also vary sharply across the wealth distribution, potentially driving wealth inequality in old age (Davies 1999). Lower-wealth retirees tend to exhaust assets more quickly, while those in the top quartile retain a significant share of their wealth into their late seventies. A notable share of retirees increase their assets during retirement, particularly among those with higher financial literacy. Using both descriptive and regression analyses, we show that financial literacy is associated with higher starting wealth, slower drawdown rates, a greater likelihood of asset accumulation, and reduced debt in later life.

These findings highlight the importance of institutional design, financial knowledge, and asset composition in shaping economic outcomes in retirement. Our results have implications for retirement policy, financial advising, and the development of insurance products that support financial well-being across the later stages of the life course.

## **2. Data and Summary Statistics**

We use data from the Health and Retirement Study (HRS), a nationally representative longitudinal survey of U.S. adults over the age of 50. The HRS collects detailed information on

wealth, income, demographics, and retirement behavior, making it well-suited for studying asset decumulation during later life.

Our analytical sample includes individuals who reached age 65 between 1995 and 2008. To accommodate the two-year survey intervals in the HRS, we also include respondents who reached age 66 or 67 within that window. We require that individuals survive at least 12 years following this age cutoff to ensure a full post-retirement observation window. This allows us to track decumulation behavior over a consistent and meaningful horizon across cohorts.

We focus on total non-housing financial wealth, which we define as the net value of the following assets: IRAs and Keogh accounts, stocks and mutual funds (including investment trusts), non-primary real estate, business holdings, checking and savings accounts (including money market funds), vehicles, certificates of deposit, bonds, and other financial assets. We subtract any non-mortgage debt to obtain net wealth. All values are adjusted to 2020 dollars using the Consumer Price Index. While we recognize housing has a distinct relationship to health and literacy (Costa-Font 2008), our wealth measure excludes housing wealth, which is less easily decumulated and typically serves both consumption and bequest functions. Throughout the analysis, we group individuals into wealth quartiles based on this non-housing financial wealth measure at age 65.

Table 1 presents summary statistics for the full sample and for each wealth quartile. The variation in asset levels is substantial. At age 65, the average retiree holds \$462,000 in non-housing financial wealth. This ranges from approximately \$3,000 in the bottom quartile to nearly \$1.4 million in the top quartile. The largest differences are observed in stock holdings, which

average \$300 in the bottom quartile and \$300,000 in the top quartile. This underscores how exposure to market-based assets is concentrated at the upper end of the wealth distribution.

Social Security wealth, estimated based on claiming at full retirement age (FRA), is more equal due to the program's progressive structure. However, disparities remain in other sources of retirement income. Pension and annuity income is highest in the top quartile and lowest in the bottom.

Demographic differences also emerge across the wealth distribution. The least wealthy quartile includes a higher share of Black household heads and fewer college-educated individuals. Educational attainment also varies widely, with the wealthiest group averaging more than 14 years of education and the least wealthy averaging around 10. It is difficult to interpret gender, as only one respondent per household is in the sample – though the higher share of female household heads in the lowest quartile coincides with a greater proportion of these households being single as well. Overall, these patterns mirror broader inequalities in lifetime earnings and savings opportunities.

We also observe sharp differences in financial literacy, a variable we define based on the standard “Big Three” questions covering compound interest, inflation, and risk diversification (Lusardi and Mitchell 2011). Among the top wealth quartile, 61 percent of individuals answered all three correctly, compared to just 21 percent in the bottom quartile. This correlation between financial literacy and wealth at retirement will be explored further in the analysis that follows.

**Table 1: Summary Statistics at Age 65 by Wealth Quartile**

|                            | (1)<br>All<br>Individuals | (2)<br>Wealth<br>Quartile 1 | (3)<br>Wealth<br>Quartile 2 | (4)<br>Wealth<br>Quartile 3 | (5)<br>Wealth<br>Quartile 4 |
|----------------------------|---------------------------|-----------------------------|-----------------------------|-----------------------------|-----------------------------|
| Total Wealth (1000s)       | 461.95<br>(1729.79)       | 3.44<br>(22.20)             | 45.26<br>(45.75)            | 178.14<br>(120.98)          | 1,390.25<br>(3006.74)       |
| Stocks (1000s)             | 91.64<br>(469.81)         | 0.30<br>(2.63)              | 3.24<br>(13.16)             | 25.49<br>(56.69)            | 289.38<br>(836.80)          |
| IRAs (1000s)               | 92.42<br>(308.76)         | 0.88<br>(5.23)              | 9.17<br>(23.36)             | 52.30<br>(81.86)            | 263.07<br>(528.10)          |
| Checking Accounts (1000s)  | 34.45<br>(102.37)         | 2.12<br>(5.25)              | 11.51<br>(20.04)            | 29.06<br>(43.09)            | 81.83<br>(174.78)           |
| Social Security Wealth     | 133,873.60<br>(86,902.63) | 97,257.10<br>(54,008.93)    | 104,025.10<br>(56,473.22)   | 110,107.30<br>(62,588.05)   | 123,061.70<br>(65,690.82)   |
| Pension and Annuity Income | 12,853.70<br>(476,098.81) | 2,695.55<br>(114,56.93)     | 5,817.15<br>(12,002.38)     | 9,260.41<br>(19,486.61)     | 10,551.33<br>(21,671.29)    |
| Birth Year                 | 1936.38<br>(4.07)         | 1937.30<br>(3.81)           | 1936.66<br>(3.76)           | 1936.60<br>(3.82)           | 1937.44<br>(3.79)           |
| Male                       | 0.50<br>(0.50)            | 0.39<br>(0.49)              | 0.51<br>(0.50)              | 0.52<br>(0.50)              | 0.62<br>(0.49)              |
| White                      | 0.86<br>(0.35)            | 0.70<br>(0.46)              | 0.82<br>(0.38)              | 0.90<br>(0.30)              | 0.95<br>(0.22)              |
| Black                      | 0.10<br>(0.30)            | 0.23<br>(0.42)              | 0.15<br>(0.35)              | 0.06<br>(0.24)              | 0.02<br>(0.15)              |
| Hispanic                   | 0.07<br>(0.26)            | 0.17<br>(0.37)              | 0.08<br>(0.27)              | 0.05<br>(0.22)              | 0.02<br>(0.14)              |
| Years of Education         | 12.64<br>(3.13)           | 10.61<br>(3.54)             | 11.90<br>(2.85)             | 12.98<br>(2.54)             | 14.27<br>(2.48)             |
| 100% Financial Literacy    | 0.41<br>(0.49)            | 0.21<br>(0.41)              | 0.27<br>(0.45)              | 0.43<br>(0.50)              | 0.61<br>(0.49)              |
| <i>N</i>                   | 36,562                    | 1,482                       | 1,506                       | 1,509                       | 1,513                       |

Notes: Values are means at age 65 (or 66–67, due to the HRS survey cycle), reported in 2020 U.S. dollars. Wealth quartiles are based on non-housing financial wealth, which excludes primary residences and adjusts for debt. Social Security wealth is estimated assuming claiming at full retirement age. The financial literacy indicator equals one if the respondent answers all three standard questions on compound interest, inflation, and risk diversification correctly. Source: HRS, 1995–2020.

### 3. Results

#### 3.1 Asset Decumulation over Time by Wealth Quartile

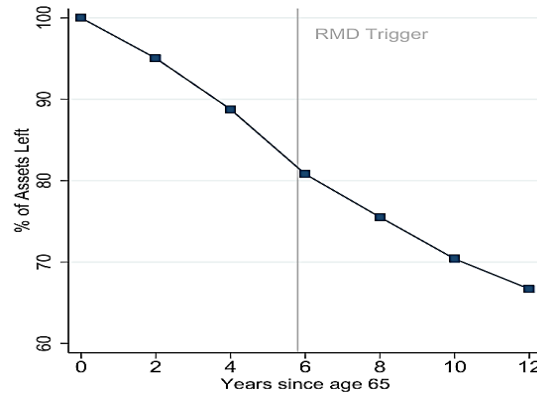
We begin by examining how much non-housing financial wealth retirees retain over time. Figure 1 shows that the median individual retains approximately 68 percent of their financial wealth 12 years after turning 65. People begin to decumulate even prior to the required minimum distribution trigger, which is shown at age 72 in the figure (it has since increased to age 75).

Appendix Figure 1 extends the analysis to include housing assets. When housing is included, the median asset retention rate rises to nearly 85 percent, reflecting the relatively slow drawdown of home equity in retirement.

To understand heterogeneity in decumulation, we next examine patterns by initial wealth quartile. Figure 2 plots the evolution of logged non-housing financial wealth over time for each quartile. These estimates are based on the following regression specification which is run for each quartile separately:

$$Y_{i,t} = \text{YearsSinceAge65}_{i,t} + \epsilon_{i,t}, \quad (1)$$

**Figure 1: Median Asset Retention Over Time**



Notes: This figure depicts the percentage of assets (wealth) remaining over time relative to the baseline in the year one turns 65. Source: HRS, 1995-2020.

where the dependent variable is the log of total non-housing financial wealth. We apply the inverse hyperbolic sine (IHS) transformation to this measure to account for the high degree of skewness and to accommodate observations with zero or negative values. For simplicity, we refer to this transformation as the ‘log’ of assets throughout the paper. We interpret coefficients in line with Bellemare and Wichman (2020).  $\text{YearsSinceAge65}_{i,t}$  represent dummy variables

for 2, 4, 6, 10, and 12 years after reaching age 65. We omit the year one turns 65 so that the coefficients are in reference to the start of the retirement-age period as a baseline.

The decumulation profiles in Figure 2 reveal several key patterns. First, wealth trajectories differ markedly by initial wealth quartile. In the bottom quartile (panel d), individuals on average begin retirement with near-zero or negative wealth, and show modest gains in the early years. This reflects debt repayment or small asset accumulation from continued income or transfers. In contrast, individuals in the second through fourth quartiles show consistent declines in wealth over time, consistent with asset decumulation.

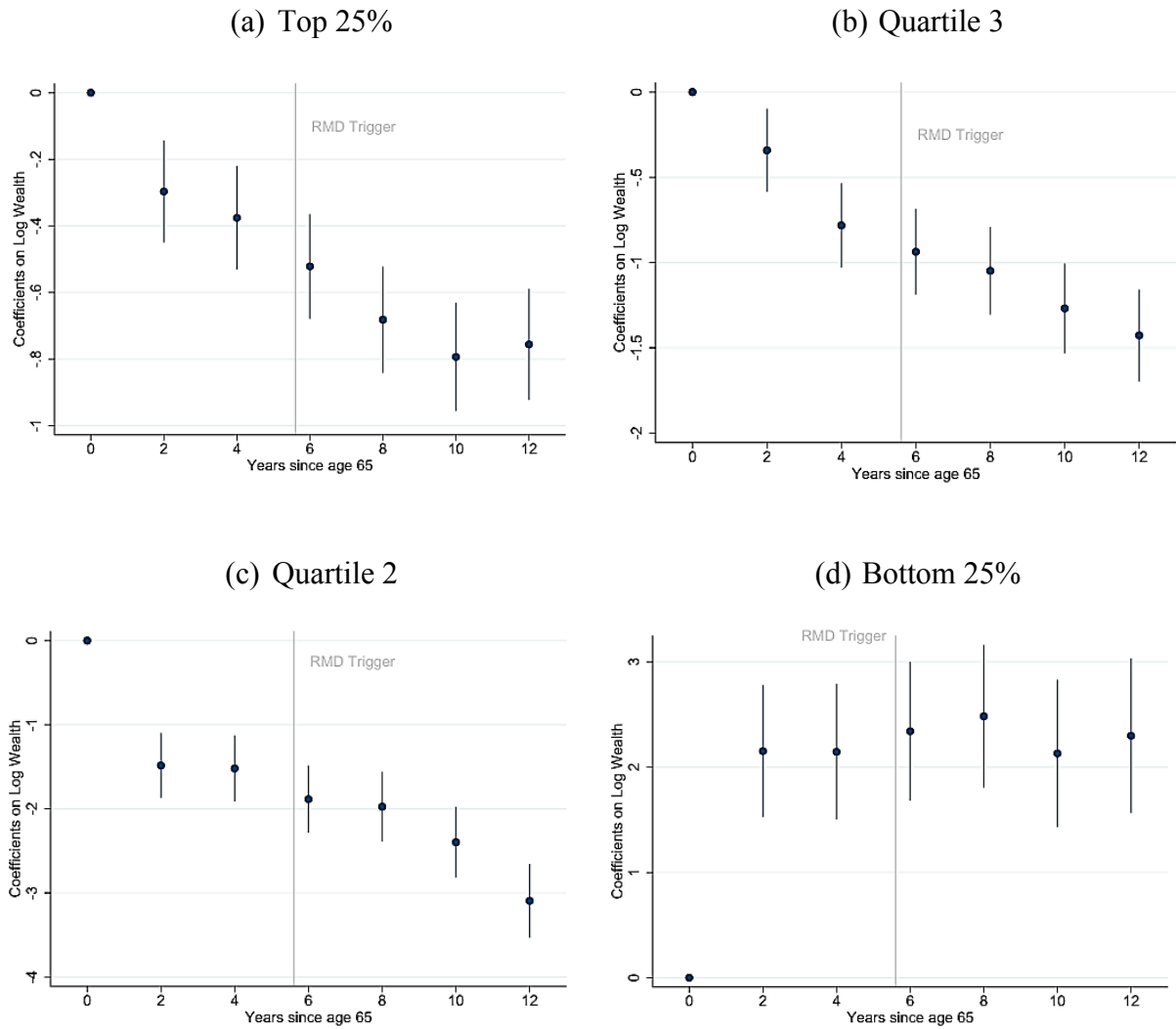
Second, the rate and extent of decumulation vary substantially across quartiles. By year 12, the mean wealth decline is 54 percent in the top quartile, 76 percent in the third, and 95 percent in the second. This pattern reflects both differences in starting wealth and variation in exposure to different asset types, which we explore in the next section.

Figure 3 complements this analysis by showing the median percentage of assets retained over time for the second through fourth quartiles. While the top quartile shows a relatively rapid drawdown early in retirement, likely reflecting IRA withdrawals around the RMD age, this group still retains roughly 71 percent of their wealth after 12 years. Median retention is lower in the third and second quartiles, at 67 percent and 55 percent, respectively. These results suggest meaningful heterogeneity in retirement decumulation patterns, even outside the extremes of the wealth distribution.

Given that the bottom quartile has limited or negative wealth at retirement, we exclude this group from most of the following analyses. This group is important for policy, however, as they arrive to older ages with little to decumulate – and in fact, sometimes significant debt.

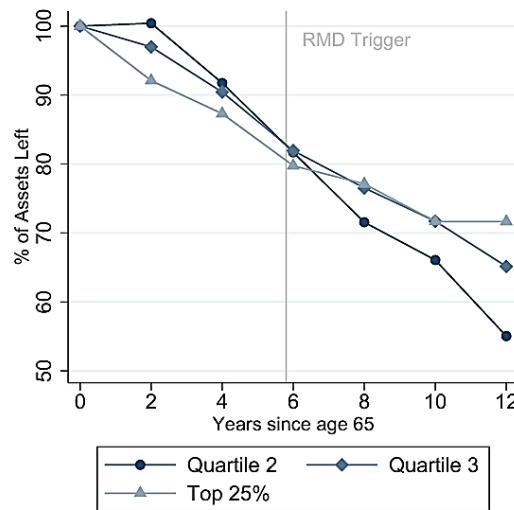


**Figure 2: Mean Log Wealth over Time by Initial Wealth Quartile**



Notes: Coefficients are from regressions of inverse hyperbolic sine–transformed non-housing financial wealth on years since age 65, with age 65 as the omitted category. Panels show separate regressions by wealth quartile at baseline. Source: HRS, 1995–2020.

**Figure 3: Median Asset Retention Over Time by Wealth Quartile**



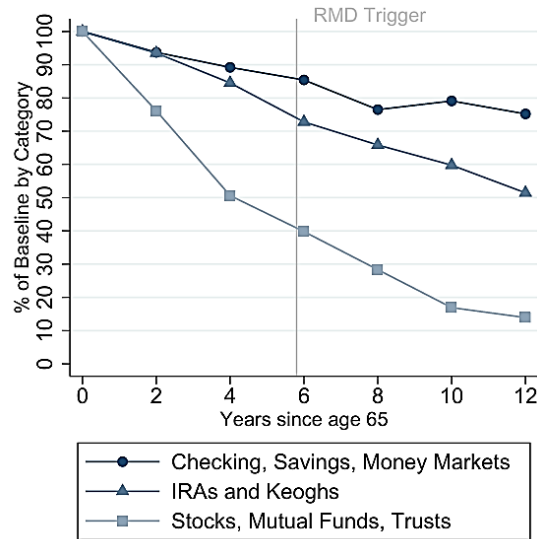
Notes: This figure depicts the percentage of non-housing financial wealth remaining over time, relative to the baseline in the year one turns 65. Results are shown separately for the second, third, and fourth wealth quartiles. Source: HRS, 1995–2020.

### ***3.2 Decumulation Patterns by Asset Type and Wealth Quartile***

To better understand what drives differences in overall wealth decumulation, we examine how drawdown patterns vary across asset types. We focus on three broad categories of non-housing financial wealth: (1) checking and savings accounts, (2) IRAs and Keogh accounts, and (3) stocks, mutual funds, and investment trusts.

Figure 4 presents median retention rates by asset type over the 12 years following age 65. Stock holdings are drawn down most aggressively, with nearly 85 percent of value decumulated by year 12. IRA balances decline more gradually, with roughly 50 percent drawn down over the same period. In contrast, checking and savings accounts show minimal decumulation. The median balance in these accounts declines by only about 25 percent, suggesting that many retirees maintain liquidity buffers even as other assets are spent down.

**Figure 4: Median Asset Retention by Asset Type**



Notes: This figure shows the median percentage of each asset type—checking accounts, IRAs, and stocks—remaining over time, relative to values at age 65. Source: HRS, 1995–2020.

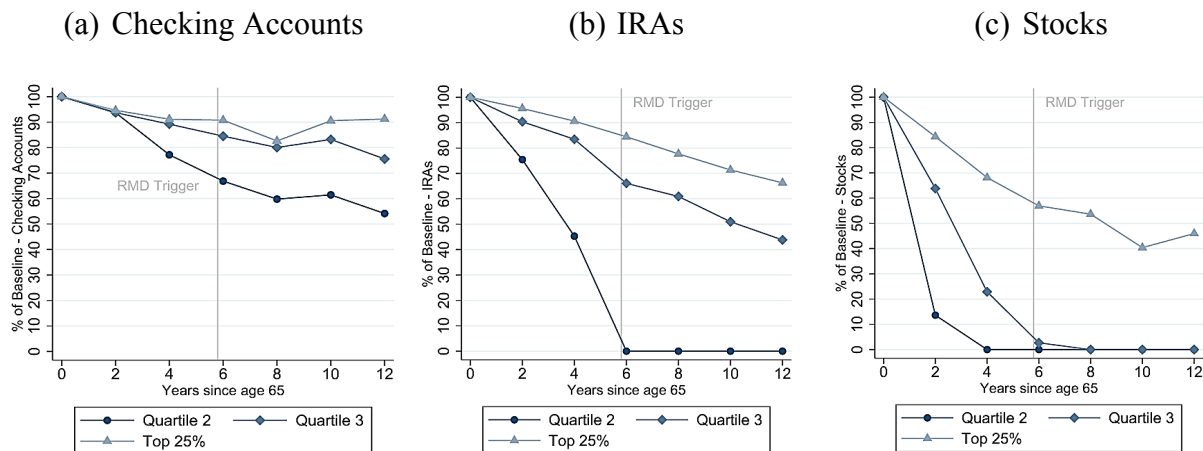
To assess how these patterns differ across the wealth distribution, Figure 5 shows asset-specific decumulation trajectories by wealth quartile (excluding the bottom quartile due to low and volatile baseline values). The sharp drawdown in stocks observed in aggregate is largely driven by the second and third quartiles, which tend to hold relatively small stock balances and appear to use them early in retirement. The top quartile decumulates stock assets more gradually and continues to hold substantial balances throughout the 12-year window.

IRA decumulation appears smoother and more consistent across the third and fourth quartiles. The timing and pace of this drawdown align with institutional features of the tax system, particularly Required Minimum Distributions (RMDs), which begin in the mid-70s. In contrast, checking and savings balances remain relatively stable in all quartiles. These accounts represent a larger share of wealth in lower quartiles but are not significantly drawn down even among higher-wealth retirees.

Figure 6 shows how these asset-specific decumulation patterns differ by cohort. We divide the sample into two groups: retirees who turned 65 between 1995 and 2000, and those who turned 65 between 2002 and 2008. The younger cohort decumulates more rapidly in the early years of retirement, while the older cohort shows a more gradual drawdown early on but converges to a similar asset level by year 12. These patterns may reflect differences in market timing, macroeconomic conditions, or behavioral preferences. While the magnitude of the cohort difference is modest, it suggests that decumulation behavior may evolve across generations and in response to external factors.

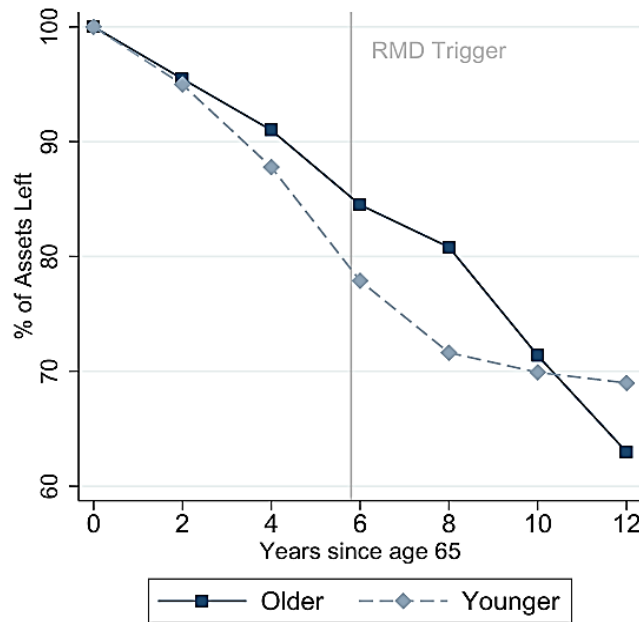
Overall, these results reinforce the importance of distinguishing asset types when analyzing decumulation. They also highlight how wealth quartile and birth cohort shape retirees' financial trajectories in ways that may inform product design, policy, and financial guidance.

**Figure 5: Asset Decumulation by Asset Type and Wealth Quartile**



Notes: This figure shows the median retention of each asset type over time, stratified by wealth quartile at age 65. The bottom quartile is excluded due to limited and volatile asset values. Source: HRS, 1995–2020.

**Figure 6: Decumulation in the Aggregate by Cohort**



Notes: This figure compares the median percentage of total non-housing financial wealth remaining over time for two cohorts: those turning 65 between 1995–2000 and those between 2002–2008. Source: HRS, 1995–2020.

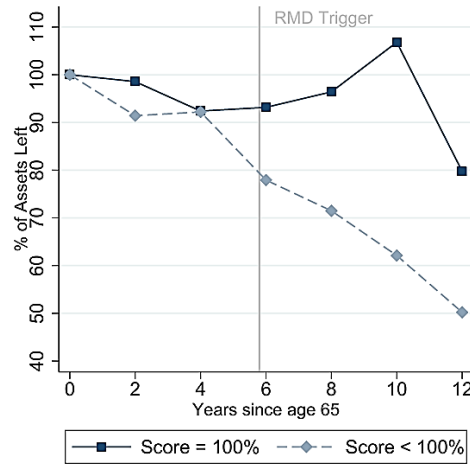
### ***3.3 Financial Literacy, Wealth Accumulation, and Debt Reduction in Retirement***

We now turn to the role of financial literacy in shaping decumulation, asset accumulation, and financial resilience in retirement. Prior research shows that financial literacy is associated with higher savings, more diversified portfolios, and less borrowing prior to retirement (Lusardi and Mitchell 2011; van Rooij, Lusardi, and Alessie 2012). We examine whether similar advantages persist after retirement, particularly in terms of wealth preservation and accumulation.

Figure 7 shows that retirees with high financial literacy begin retirement with substantially more non-housing financial wealth than their less literate counterparts, and that this gap widens over time. High-literacy individuals retain more wealth throughout the 12-year period and decumulate at a slower pace. Median wealth for the high-literacy group falls by about

23 percent, compared to 45 percent for the low-literacy group. Notably, a sizable share of retirees with high financial literacy accumulate additional wealth during retirement, offsetting declines observed elsewhere in the distribution.

**Figure 7: Median Wealth Over Time by Financial Literacy**



Notes: This figure shows the median level of non-housing financial wealth over time, relative to baseline at age 65, for individuals with high versus low financial literacy. High literacy is defined as answering all three standard financial literacy questions correctly. Source: HRS, 1995–2020.

Next we ask investigate these relationships in a regression setting. Namely, if we control for wealth quartile and years in after age 65, does financial literacy (still) relate positively to wealth?

We can consider this an estimation within-quartile and controlling for time. We estimate regressions of the following form:

$$Y_{i,t} = HighFL_i + WealthQ_i + (WealthQ_i * YearsSinceAge65_{i,t}) + \epsilon_{i,t}, \quad (2)$$

where the dependent variable is the logged version of total non-housing financial wealth, stock wealth, checking account wealth, IRA wealth, or other wealth.  $HighFL_i$  is a dummy variable

equal to one if an individual answers all three literacy questions correctly (i.e., higher financial literacy) and zero otherwise.

Table 2 reports the regression results. Financial literacy is strongly associated with the likelihood of wealth accumulation. Individuals with high financial literacy are 12 percentage points more likely to increase their assets during retirement, conditional on observable characteristics. This result suggests that financial literacy continues to shape outcomes well beyond the point of retirement entry.

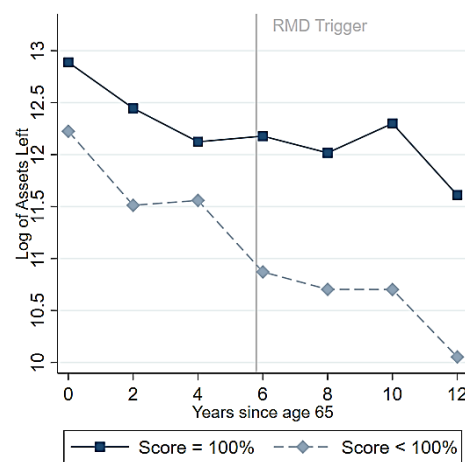
**Table 2: Predictors of Wealth Accumulation After Age 65**

|                           | (1) Log<br>Wealth    | (2) Log<br>Stock<br>Wealth | (3) Log<br>Checking<br>Wealth | (4) Log<br>IRA<br>Wealth | (5) Log<br>Other<br>Wealth |
|---------------------------|----------------------|----------------------------|-------------------------------|--------------------------|----------------------------|
| Financial Literacy 100%   | 0.607*<br>(0.312)    | 0.668<br>(0.417)           | 0.348<br>(0.263)              | 1.134**<br>(0.489)       | 0.405*<br>(0.229)          |
| Wealth Quartile 2 (Q2)    | 6.689***<br>(0.771)  | 0.694**<br>(0.345)         | 3.163***<br>(0.487)           | 1.736***<br>(0.480)      | 3.236***<br>(0.477)        |
| Wealth Quartile 3 (Q3)    | 8.642***<br>(0.742)  | 3.686***<br>(0.563)        | 5.288***<br>(0.437)           | 6.241***<br>(0.608)      | 4.429***<br>(0.450)        |
| Wealth Quartile 4 (Q4)    | 9.900***<br>(0.751)  | 8.068***<br>(0.721)        | 5.981***<br>(0.466)           | 8.937***<br>(0.653)      | 5.206***<br>(0.486)        |
| Years Since Age 65 (Time) | 0.126<br>(0.095)     | -0.025<br>(0.021)          | -0.029<br>(0.049)             | -0.003<br>(0.031)        | -0.057<br>(0.057)          |
| Q2*Time                   | -0.370***<br>(0.110) | 0.003<br>(0.041)           | -0.049<br>(0.061)             | -0.075<br>(0.055)        | -0.108<br>(0.067)          |
| Q3*Time                   | -0.214**<br>(0.103)  | -0.094*<br>(0.052)         | -0.045<br>(0.058)             | -0.117*<br>(0.065)       | -0.025<br>(0.061)          |
| Q4*Time                   | -0.148<br>(0.097)    | -0.010<br>(0.072)          | -0.050<br>(0.062)             | -0.030<br>(0.066)        | -0.003<br>(0.064)          |
| Observations              | 2,139                | 2,139                      | 2,139                         | 2,139                    | 2,139                      |
| R <sup>2</sup>            | 0.331                | 0.328                      | 0.284                         | 0.362                    | 0.308                      |

Notes: This table depicts results of estimating Equation (2) with dependent variables as listed in the column headings. Wealth is measured in 2020 US dollars, and the inverse hyperbolic sine function was used to transform it. Standard errors are clustered at the individual level. Sources: HRS, 1995-2020. Source: HRS, 1995–2020. Significance is given by: \*  $p < 0.10$ , \*\*  $p < 0.05$ , and \*\*\*  $p < 0.01$ .

As a robustness check, Figure 8 plots wealth trajectories using an inverse hyperbolic sine transformation of non-housing financial wealth. The divergence between high- and low-literacy groups remains evident, reinforcing the earlier findings and suggesting that the results are not driven by extreme values or skewness in the wealth distribution.

**Figure 8: Log Wealth Over Time by Financial Literacy**



Notes: This figure shows the inverse hyperbolic sine (IHS) transformation of non-housing financial wealth over time, separately for individuals with high versus low financial literacy. High literacy is defined as answering all three standard financial literacy questions correctly. Source: HRS, 1995–2020.

While we do not directly analyze household debt, the broader literature suggests that retirees with low financial literacy often face ongoing financial vulnerabilities. Lusardi, Mitchell, and Oggero (2020) find that debt burdens persist into retirement and are associated with greater economic fragility. In this context, our findings suggest that financial literacy not only supports asset preservation but may also protect against other forms of financial strain during later life.

Together, these results highlight the importance of financial literacy for retirement security. Literacy is associated with slower drawdown of financial assets, a greater likelihood of



net asset accumulation, and continued financial stability through retirement. These effects persist even after adjusting for education and initial wealth, underscoring the value of financial capability as a distinct resource. Policies and interventions that support financial literacy could help narrow retirement preparedness gaps and reduce vulnerability among older adults.

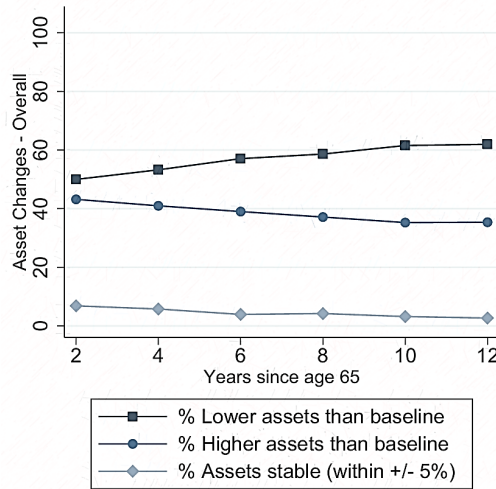
### ***3.4 Distinguishing Accumulators, Decumulators, and Asset Holders***

The aggregate patterns shown earlier obscure important heterogeneity in how individuals manage their financial assets in retirement. While some households draw down their wealth steadily, others maintain stable asset levels or even accumulate wealth over time. In this section, we distinguish among these groups and examine the characteristics that predict these divergent retirement trajectories. We define three mutually exclusive categories based on changes in non-housing financial wealth over the 12 years following age 65:

- (1) Accumulators, whose wealth increases by more than 5 percent;
- (2) Decumulators, whose wealth declines by more than 5 percent; and
- (3) Asset Holders, whose wealth remains within  $\pm 5$  percent of baseline.

We next study the evolution of these groups over time after reaching age 65. Figure 9 suggests, in aggregate, very few individuals are to hold their asset levels steady at older ages. Most people, experience decreasing assets over time. Nearly 40% of the retirees, however, are increasing their assets over the 12-year period following age 65.

**Figure 9: Retirement Wealth Trajectories**

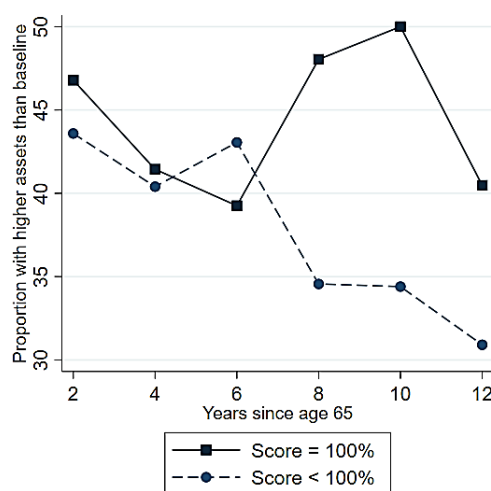


Notes: This figure shows the share of individuals classified as accumulators, decumulators, or asset holders by baseline wealth quartile at age 65. Asset holders are defined as those whose financial wealth remains within  $\pm 5$  percent of baseline after 12 years. Source: HRS, 1995–2020.

Figure 10 presents the same breakdown by financial literacy. Retirees with high financial literacy are significantly more likely to be accumulators. These patterns are consistent with prior results showing that financial literacy supports slower drawdown and greater asset preservation. A question that emerges from these descriptives is how the literacy-decumulation relationship varies across wealth types. In Table 3, we present results from estimating Equation (2) but with a new set of dependent variables. Namely, we use dependent variables that are dummy variables equal to one if wealth in a given category is equal to, or above, the baseline from the year of turning age 65. After controlling, again, for wealth quartile and years retired, high financial literacy relates to significant asset accumulation during retirement in terms of both total non-housing financial wealth and in terms of other wealth (such as vehicles, businesses, certificates of deposits, and bonds). In fact, having a perfect score on the three financial literacy questions implies an 8.6% greater chance of increasing overall wealth levels in retirement than those who do not receive a perfect score. There is also a new finding here in column (5), which is that high

financial literacy is related to significantly less debt over the course of retirement, relative to those of low financial literacy. Specifically, having a perfect literacy score implies an 11% reduction in the chance of having the same or more debt in retirement (relative to the year one turned 65). This has important implications for bequests and other financial decisions in old age.

**Figure 10: Retirement Wealth Trajectories by Financial Literacy**



Notes: This figure shows the distribution of accumulators, decumulators, and asset holders by financial literacy status. High literacy is defined as answering all three standard financial literacy questions correctly. Asset holders are those whose financial wealth remains within  $\pm 5$  percent of baseline after 12 years. Source: HRS, 1995–2020.

**Table 3: Predictors of Wealth Accumulation and Stability**

|                           | (1)<br>Overall       | (2)<br>Stocks        | (3)<br>IRAs          | (4)<br>Checking<br>Accounts | (5)<br>Debt          | (6)<br>Other<br>Assets |
|---------------------------|----------------------|----------------------|----------------------|-----------------------------|----------------------|------------------------|
| Financial Literacy 100%   | 0.086**<br>(0.034)   | -0.027<br>(0.034)    | -0.027<br>(0.035)    | 0.029<br>(0.035)            | -0.114***<br>(0.036) | 0.071**<br>(0.035)     |
| Wealth Quartile 2 (Q2)    | -0.070*<br>(0.042)   | -0.025<br>(0.018)    | -0.036<br>(0.023)    | -0.018<br>(0.044)           | 0.096***<br>(0.036)  | -0.132***<br>(0.042)   |
| Wealth Quartile 3 (Q3)    | -0.016<br>(0.040)    | -0.078***<br>(0.025) | -0.153***<br>(0.029) | -0.004<br>(0.044)           | 0.136***<br>(0.034)  | -0.081*<br>(0.042)     |
| Wealth Quartile 4 (Q4)    | -0.154***<br>(0.044) | -0.225***<br>(0.033) | -0.214***<br>(0.038) | -0.020<br>(0.049)           | 0.151***<br>(0.038)  | -0.161***<br>(0.045)   |
| Years Since Age 65 (Time) | -0.021***<br>(0.006) | -0.003<br>(0.002)    | -0.003<br>(0.002)    | -0.028***<br>(0.006)        | -0.037***<br>(0.006) | -0.037***<br>(0.006)   |
| Q2*Time                   | -0.017**<br>(0.008)  | -0.002<br>(0.003)    | -0.008*<br>(0.004)   | -0.005<br>(0.007)           | 0.025***<br>(0.006)  | -0.002<br>(0.007)      |
| Q3*Time                   | -0.018**<br>(0.007)  | -0.017***<br>(0.005) | -0.025***<br>(0.005) | -0.002<br>(0.008)           | 0.025***<br>(0.006)  | -0.000<br>(0.007)      |
| Q4*Time                   | -0.016**<br>(0.008)  | -0.021***<br>(0.006) | -0.031***<br>(0.005) | -0.010<br>(0.008)           | 0.033***<br>(0.006)  | 0.001<br>(0.008)       |
| Observations              | 2,139                | 2,139                | 2,139                | 2,139                       | 2,139                | 2,139                  |
| R <sup>2</sup>            | 0.106                | 0.149                | 0.176                | 0.071                       | 0.124                | 0.102                  |

Notes: This table depicts results of estimating Equation (2) with dependent variables set as dummy variables equal to one if an account's (given by the column heading) value is the same or greater than the baseline value in the one turns 65. Standard errors are clustered at the individual level. Sources: HRS, 1995-2020. Significance is given by: \*  $p < 0.10$ , \*\*  $p < 0.05$ , and \*\*\*  $p < 0.01$ .

These findings challenge the assumption that decumulation is a universal or dominant pattern among retirees. Many households preserve or even grow their financial assets after retirement. These behaviors may reflect a variety of motives, including precaution, uncertainty about longevity or medical expenses, or intentional bequest planning. The consistent association between financial literacy and more stable or upward financial trajectories suggests that improving decision-making capacity could play a role in strengthening retirement security.

#### **4. Conclusion**

This paper examines patterns of financial asset decumulation in retirement using longitudinal data from the Health and Retirement Study. By following individuals age 65, we characterize how non-housing financial wealth evolves over a 12-year horizon, with attention to heterogeneity by baseline wealth, asset type, and financial literacy. These findings are important in light of recent work on how late-life wealth is predictive of many outcomes including work patterns and healthy life expectancy (Bavafa, Mukherjee, and Welch 2023).

We document three main findings. First, while many households experience some asset drawdown after age 65, the pace and extent of decumulation vary widely. A substantial share of retirees maintain or even increase their financial wealth over time, particularly those in the top wealth quartile. These patterns deviate from canonical life-cycle models, which typically predict smooth and continuous decumulation during retirement. Second, decumulation differs markedly across asset types. IRA balances decline gradually, consistent with the influence of required minimum distributions and tax-deferred treatment. Stock holdings fall more sharply, especially among middle-wealth households, while checking and savings balances remain relatively stable. These results suggest that retirees segment their portfolios and draw differentially based on liquidity, tax treatment, and perceived function.

Third, financial literacy is strongly associated with wealth trajectories in retirement. Not only do the most financially literate have higher asset levels across the board, but they are also slower to decumulate and much more likely to accumulate wealth (and decrease debt levels). To the extent to which decumulation patterns are a driver of inequality in old age, potentially in combination with (healthy) life expectancy, it appears improving financial literacy may be one

way to support individuals' ability to slowly draw down wealth over time and, potentially, even improve financial status later in life. Taken together, our findings underscore the heterogeneity of retirement financial behavior and the importance of examining asset composition in decumulation.

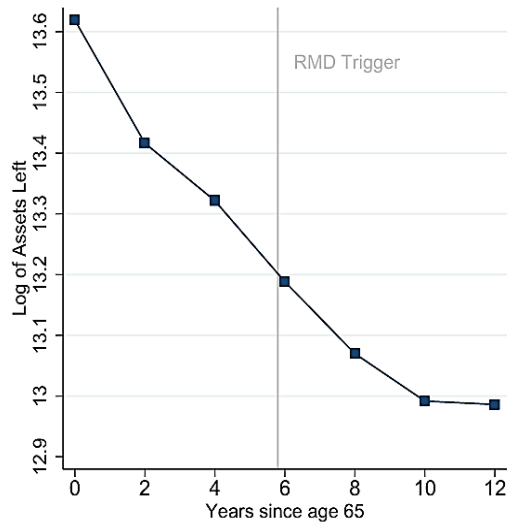
## References

- Banks, J., Blundell, R., & Tanner, S. (1998). Is There a Retirement-Savings Puzzle? *American Economic Review*, 88(4), 769-788.
- Bavafa, H., Mukherjee, A., & Welch, T. Q. (2023). Inequality in the golden years: Wealth gradients in disability-free and work-free longevity in the United States. *Journal of Health Economics*, 102820.
- Bellemare, M., & Wichman, C. J. (2020). Elasticities and the Inverse Hyperbolic Sine Transformation. *Oxford Bulletin of Economics and Statistics*, 82(1), 50-61.
- Brady, P. J., & Boguth, O. (2023). Retirement wealth decumulation and its implications for annuitization. *Journal of Pension Economics and Finance*, 22(2), 238–258.
- Brown, J. R., Poterba, J. M., & Richardson, D. P. (2025). Trends in retirement and retirement income choices by TIAA participants: 2000–2018. *Journal of Pension Economics & Finance*, 24, 47-68.
- Brown, J. R. (2009). Understanding the Role of Annuities in Retirement Planning. In: *Overcoming the Saving Slump: How to Increase the Effectiveness of Financial Education and Saving Programs* (Annamaria Lusardi, ed.), 178-206.
- Christensen, B. J., Kallestrup-Lamb, M., & Kennan, J. (2022). *Consumption and Saving After Retirement*. Working Paper, NBER #29826.
- Costa-Font, J. (2008). Housing assets and the socio-economic determinants of health and disability in old age. *Health & Place*, 14, 478-491.
- Davies, J.B. (1999). Age, Wealth Inequality, and Life-Cycle Modelling. *Geneva Papers on Risk and Insurance* 24(1), 64-76.
- Davis, B., Leive, A., & Gellert, A. (2023). Fungibility in workplace benefits choices: Evidence from Health Savings Accounts. *TIAA Institute Research Paper Series*.
- De Nardi, M., French, E., & Jones, J. B. (2016). Savings after retirement: A survey. *Annual Review of Economics*, 8, 177–204.

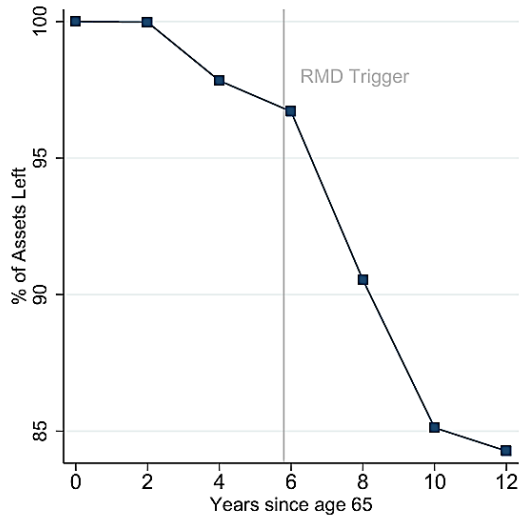
- De Nardi, M., French, E., Jones, J. B., & McGee, R. (2025). Why do Couples and Singles Save during Retirement? Household Heterogeneity and Its Aggregate Implications. *Journal of Political Economy*, 133(3), 750-792.
- Goda, G. S., Levy, M., Manchester, C. F., Sojourner, A., & Tasoff, J. (2019). Predicting Retirement Savings Using Survey Measures of Exponential-Growth Bias and Present Bias. *Economic Inquiry*, 57(3), 1636-1658.
- Goodman, L., Mukherjee, A., & Ramnath, S. (2023). Set it and Forget it? Financing Retirement in an Age of Defaults. *Journal of Financial Economics*, 148(1), 47-68.
- Hurd, M. D., & Rohwedder, S. (2008). *The Retirement Consumption Puzzle: Actual Spending Change in Panel Data*. Working Paper, NBER #13929.
- Kieren, P., & Weber, M. (2022). When saving is not enough—wealth decumulation in retirement. *Journal of Pension Economics & Finance*, 21(3), 446-473.
- Lockwood, L. M. (2012). Bequest motives and the annuity puzzle. *Review of Economic Dynamics*, 15(2), 226-243.
- Lusardi, A. & Mitchell, O.S. (2011). Financial literacy around the world: an overview. *Journal of Pension Economics and Finance*, 10(4), 497-508.
- Lusardi, A., Mitchell, O. S., & Oggero, N. (2020). Debt and financial vulnerability on the verge of retirement. *Journal of Money, Credit and Banking*, 52(1), 100–118.
- Olafsson, A., & Pagel, M. (2018). Retirement puzzles: New evidence from personal finances. *Journal of Public Economics*, 234, 105013.
- Perun, P., & Yakoboski, P. J. (2014). Aging and America: Demographic Change and Its Consequences for Work and Retirement. *TIAA-CREF Institute*.
- Poterba, J., Venti, S., & Wise, D. (2011). The Composition and Drawdown of Wealth in Retirement. *Journal of Economic Perspectives*, 25(4), 95-118.
- Yaari, M. E. (1965). Uncertain lifetime, life insurance, and the theory of the consumer. *The Review of Economic Studies*, 32(2), 137-150.
- Yakoboski, P., Lusardi, A., & Hasler, A. (2023). Financial literacy, longevity literacy, and retirement readiness. *TIAA Institute Research Paper Series*.

## Appendix Figure 1: Decumulation of Total Wealth (including housing)

(a) Log of Wealth



(b) Percentage of Wealth Left



Notes: The left figure depicts the log of mean total assets (wealth), including housing, remaining by years since age 65. This value is measured in 2020 US dollars. The right figure does the same but for the median percentage of assets (wealth), including housing, remaining relative to the baseline in the year turning 65. Source: HRS, 1995-2020.